

HVAC Load Calculation

Heat Gains | CLTD Method | Worked Example | System Selection

What You Will Learn

- 7 sources of cooling load: solar, transmission, people, lighting, equipment, infiltration, ventilation
- CLTD/CLF method: the most widely used simplified load calculation approach
- Key formulas for each heat gain type with correct units and multipliers
- Complete worked example: 6×6m office room load calculation step by step

Transmission Load (walls, roof, floor)

$$Q = U \times A \times CLTD$$

U = U-value (W/m²·K) A = area (m²) CLTD = cooling load temp diff (K)

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HVAC-010: HVAC Load Calculation

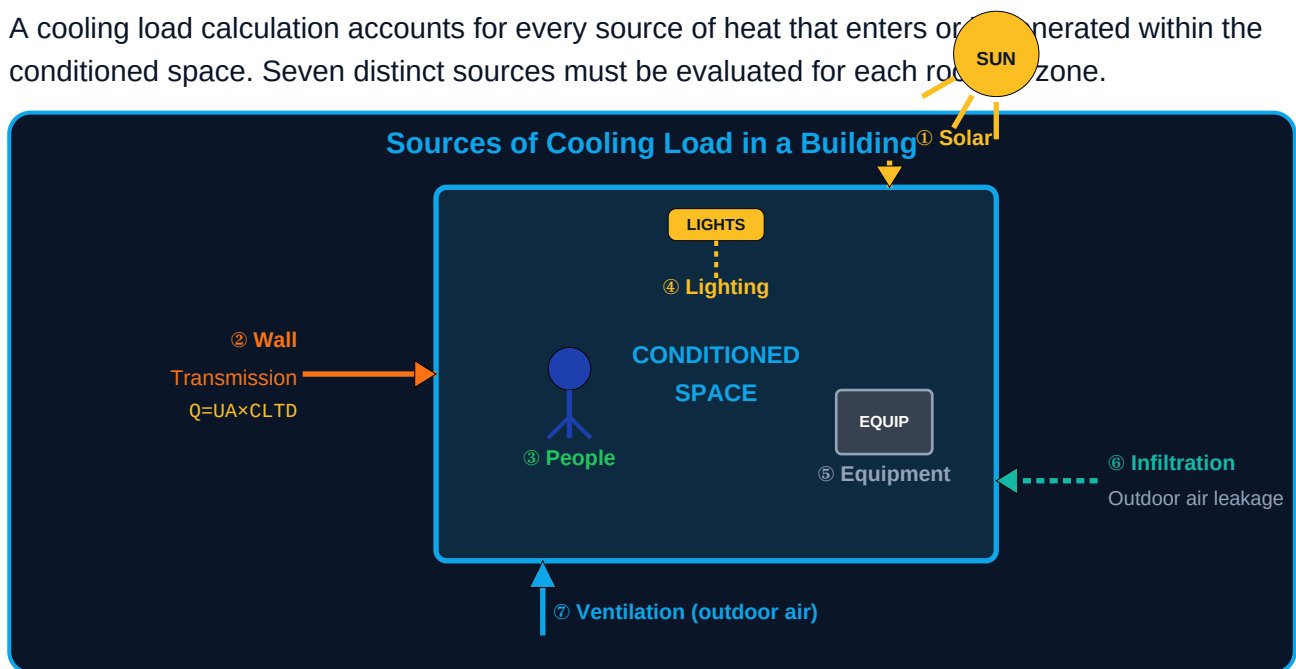
An HVAC load calculation determines how much cooling or heating power a building space requires to maintain comfort conditions. It is the essential first step in selecting any air conditioning system — without it, equipment is guessed rather than sized, leading to under-cooling (occupant discomfort), over-cooling (wasted energy and poor humidity control), or incorrect duct and pipe sizing. Every professional HVAC engineer must perform this calculation from first principles.

Learning Objectives

- Identify all seven sources of cooling load in a conditioned space
- Explain the CLTD/CLF simplified calculation method and when to apply it
- Apply the correct formula for each heat gain type: transmission, solar, people, lighting, equipment, infiltration, and ventilation
- Complete a full room cooling load calculation step by step using a worked example
- Select the correct air conditioning system capacity based on calculated total load

1. Sources of Cooling Load

A cooling load calculation accounts for every source of heat that enters or is generated within the conditioned space. Seven distinct sources must be evaluated for each room or zone.

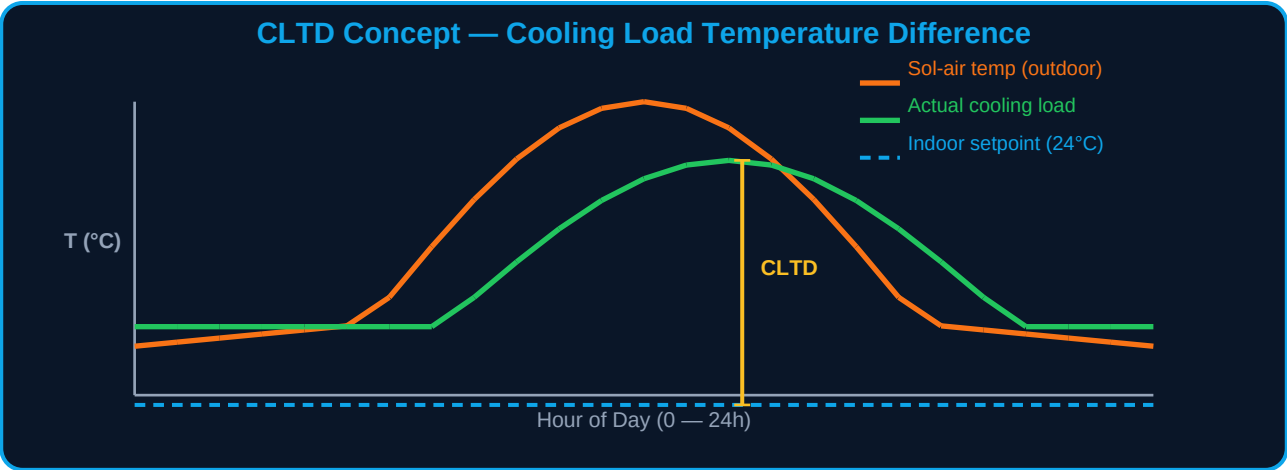


Source	What it represents	Primary Variables
① Solar (glass)	Sunlight passing through windows — the largest variable load	Window area (m^2), orientation (N/S/E/W), shading coefficient (SC), hour of day

② Trans mission	Heat conducted through walls, roof, floor, glass from outdoor air temperature difference	U-value ($\text{W/m}^2\cdot\text{K}$), area (m^2), CLTD (K)
③ People (occupants)	Metabolic heat released by people — both sensible (dry heat) and latent (moisture)	Number of people, activity level, hours of occupancy
④ Lighting	Heat from light fixtures — fluorescent, LED, and incandescent all release heat	Installed watts, type of fixture, hours of use
⑤ Equipment	Computers, printers, kitchen equipment, motors — anything that uses electricity or fuel releases heat	Equipment wattage, usage factor, heat dissipation rate
⑥ Infiltration	Unconditioned outdoor air leaking in through gaps, cracks, and poorly sealed openings	Ventilation rate (ACH), room volume, indoor/outdoor temperature and humidity difference
⑦ Ventilation	Controlled outdoor air deliberately brought in for air quality — must be cooled and dehumidified	Outside air flow rate (L/s or CFM), outside conditions vs inside setpoint

2. The CLTD/CLF Method — Simplified Load Calculation

The **Cooling Load Temperature Difference / Cooling Load Factor (CLTD/CLF)** method, published in ASHRAE Fundamentals, is the most widely used simplified approach for manual load calculations. It accounts for the thermal mass effect of walls and roofs — the fact that heat absorbed by a heavy wall during the day reaches the interior hours later (time lag) and at a reduced amplitude (damping).



Key CLTD Definitions

Term	Definition	How to Obtain
CLTD (K)	Cooling Load Temperature Difference — an equivalent temperature difference that accounts for both outdoor-indoor temperature difference and the time lag and damping of thermal mass	ASHRAE tables, looked up by wall/roof group (A–G), latitude, month, and hour
CLF	Cooling Load Factor — a factor (0 to 1) that accounts for how much of the instantaneous heat gain from internal sources (lights, people, equipment) actually becomes a cooling load at that hour	ASHRAE tables, looked up by heat source type and hours of operation
SC (Shading Coefficient)	Ratio of solar heat gain through a window compared to 3mm clear glass. SC=1.0 for clear glass; SC≈0.25 for low-e double glazing	Window manufacturer data or ASHRAE glass tables
SHGF (W/m²)	Solar Heat Gain Factor — the maximum instantaneous solar radiation through 3mm clear glass for a given orientation, latitude, and month	ASHRAE solar tables, CIBSE Guide A

3. Calculation Formulas for Each Heat Source

Heat Source	Formula	Units & Notes
Transmission (walls, roof, floor)	$Q_{trans} = U \times A \times CLTD$	$U = W/m^2 \cdot K$; $A = m^2$; CLTD from ASHRAE tables (K). Apply CLTD correction for indoor $T \neq 25^\circ C$ or outdoor mean $T \neq 29^\circ C$.
Solar gain (windows)	$Q_{solar} = A \times SC \times SHGF \times CLF$	$A =$ glazing area (m^2); $SC =$ shading coefficient; $SHGF = W/m^2$ from tables; $CLF = 0-1$ factor.
People (sensible)	$Q_{people_s} = N \times q_s \times CLF$	$N =$ number of people; $q_s =$ sensible heat per person (W) from ASHRAE table by activity; CLF from table.
People (latent)	$Q_{people_l} = N \times q_l$	$q_l =$ latent heat per person (W) from ASHRAE table. No CLF applied to latent gains.
Lighting	$Q_{light} = W_{installed} \times F_{use} \times F_{ballast} \times CLF$	$W_{installed} =$ total watts; $F_{use} =$ usage factor (0–1); $F_{ballast} = 1.0$ LED/1.25 fluorescent with magnetic ballast.
Equipment / Appliances	$Q_{equip} = W_{nameplate} \times F_{use} \times F_{load}$	$F_{use} =$ fraction of time in use; $F_{load} =$ fraction of rated load. PC: ~65W sensible heat typical.
Infiltration (sensible)	$Q_{inf_s} = 0.33 \times V_{dot} \times (T_{out} - T_{in})$	$V_{dot} =$ infiltration air flow rate (L/s); T in $^\circ C$. Factor $0.33 = \rho \times c_p / 1000$ for SI units.
Ventilation (sensible)	$Q_{vent_s} = 0.33 \times Q_{oa} \times (T_{out} - T_{in})$	$Q_{oa} =$ outside air flow rate (L/s) per ASHRAE 62.1 minimum requirements.
Ventilation (latent)	$Q_{vent_l} = 0.8 \times Q_{oa} \times (W_{out} - W_{in})$	$W =$ humidity ratio (g/kg). Factor $0.8 = \rho \times h_{fg} / 1000$ for SI. This is often the largest latent load.

4. Worked Example — Office Room Load Calculation

Scenario: A 6 m × 6 m office room, 3 m ceiling height, in Addis Ababa, Ethiopia (approx. 9°N). Calculation for 3:00 PM on a July day. Indoor setpoint: 24°C, 55% RH. Outdoor: 32°C dry bulb, 40% RH. One exterior wall (south-facing), one south-facing window (2 m × 1.5 m, single clear glass, no external shading).

Step 1 — Room Geometry

Surface	Dimension	Area (m ²)	Notes
South wall (gross)	6 m × 3 m	18.0	Brick/block wall, medium weight
South window	2 m × 1.5 m	3.0	Single clear glass, aluminium frame
South wall (net)	18.0 – 3.0	15.0	Solid wall area after deducting window
Floor / Ceiling	6 m × 6 m	36.0	Flat concrete roof (exposed to sun)
Internal walls (×3)	Each: 6 m × 3 m	54.0 total	Internal — no transmission load

Step 2 — Transmission Load (Wall + Roof)

Surface	U-value (W/m ² ·K)	Area (m ²)	CLTD (K)	Q (W)
South wall (brick 220mm)	0.85	15.0	14 (from table, 3pm, south)	$0.85 \times 15 \times 14 = \mathbf{179}$
South window (glass)	5.7	3.0	8 (glass: use outdoor-indoor diff)	$5.7 \times 3.0 \times 8 = \mathbf{137}$
Concrete roof (exposed)	1.10	36.0	28 (heavy roof, peak ~4pm)	$1.10 \times 36 \times 28 = \mathbf{1,109}$
Floor (over conditioned)	—	36.0	Negligible	0
TOTAL Transmission				1,425 W

Step 3 — Solar Gain through Window

Parameter	Value	Source
Window area (A)	3.0 m ²	From Step 1
Shading coefficient (SC)	0.86	Single clear glass (ASHRAE Table)
SHGF (south, 9°N, July, 3pm)	220 W/m ²	ASHRAE Solar Table — south orientation reduces solar in summer at low latitudes
CLF (glass, 3pm, heavy construction)	0.73	ASHRAE CLF table for south glass
$Q_{\text{solar}} = A \times SC \times SHGF \times CLF$	$3.0 \times 0.86 \times 220 \times 0.73$	= 417 W

Step 4 — Internal Heat Gains (People, Lighting, Equipment)

Source	Count/Watts	Sensible (W)	Latent (W)	Notes
People (office work)	4 persons	$4 \times 65 \times 0.85 = 221$	$4 \times 55 = 220$	$q_s=65W$, $q_l=55W$ per person for office activity; CLF=0.85 for 8h occupancy
LED lighting	200 W installed	$200 \times 0.95 \times 1.0 \times 0.85 = 161$	0	$F_{\text{use}}=0.95$, LED $F_{\text{ballast}}=1.0$, CLF=0.85
Computers (×4)	$4 \times 80W = 320W$	$320 \times 0.9 \times 0.8 = 230$	0	$F_{\text{use}}=0.9$, $F_{\text{load}}=0.8$ (not all fully loaded simultaneously)
TOTAL internal		612 W (sensible)	220 W (latent)	

Step 5 — Ventilation Load (Outside Air)

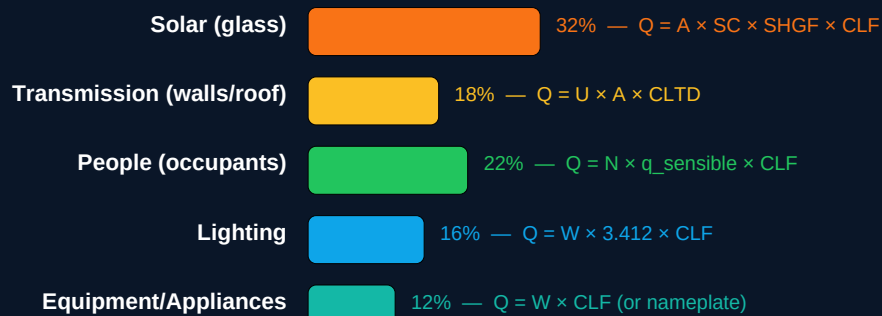
ASHRAE 62.1 minimum outside air for offices: **10 L/s per person + 0.3 L/s/m²**. For 4 people in 36 m²: $Q_{oa} = (4 \times 10) + (36 \times 0.3) = 40 + 11 = 51 \text{ L/s}$

Load Component	Formula	Calculation	Result
Ventilation sensible	$Q_{vent_s} = 0.33 \times Q_{oa} \times \Delta T$	$0.33 \times 51 \times (32-24)$	134 W
Ventilation latent	$Q_{vent_l} = 0.8 \times Q_{oa} \times \Delta W$	$0.8 \times 51 \times (14.5-10.5)^*$	163 W
Infiltration sensible	$Q_{inf_s} = 0.33 \times V_{dot} \times \Delta T$	Assume 0.5 ACH: $0.33 \times (36 \times 3 \times 0.5 / 3.6) \times 8$	44 W

* ΔW = outdoor humidity ratio (14.5 g/kg at 32°C, 40%RH) minus indoor (10.5 g/kg at 24°C, 55%RH)

Step 6 — Summary and Total Cooling Load

Typical Office Room — Cooling Load Breakdown



Total sensible cooling load = Sum of all components above

Add latent load (moisture from people + ventilation) → Room Total Load

Load Category	Sensible (W)	Latent (W)	Total (W)
Transmission (wall + roof + glass)	1,425	0	1,425
Solar gain (south window)	417	0	417
People	221	220	441
Lighting (LED)	161	0	161
Equipment (computers)	230	0	230
Ventilation (outside air)	134	163	297

Infiltration	44	0	44
ROOM TOTAL	2,632 W	383 W	3,015 W
+ Safety factor (10%)			3,317 W
Design capacity required			≈ 3.5 kW (1 TR)

Step 7 — Equipment Selection

Calculated Load	Recommended System	Capacity to Select	Notes
3.5 kW (1.0 TR)	Wall-mounted split AC unit (inverter type)	3.5 kW (12,000 BTU/h)	Select the next standard size above calculated load. Avoid over-sizing by more than 25% — oversized units short-cycle and fail to dehumidify.
3.5–7 kW (1–2 TR)	Floor-standing or ceiling-cassette split	5.0 kW (18,000 BTU/h)	Good distribution for 6×6m room. Cassette provides 4-way airflow.
7–14 kW (2–4 TR)	Ducted split or small VRF indoor unit	8–14 kW	Required when ceiling plenum available and ductwork acceptable.
>14 kW (>4 TR)	VRF / chilled water fan coil unit	Per calculation	Multiple rooms served by central plant. Requires MEP coordination.

5. Practice Exercises

#	Exercise	Data Given
1	Transmission load calculation	North-facing wall: $U=0.70 \text{ W/m}^2\cdot\text{K}$, area= 12 m^2 , CLTD= 10K . Roof: $U=1.20 \text{ W/m}^2\cdot\text{K}$, area= 25 m^2 , CLTD= 30K . Calculate Q_{trans} for each and total.
2	Solar gain calculation	West window: area= 4.0 m^2 , SC= 0.65 , SHGF= 480 W/m^2 (west, 4pm), CLF= 0.82 . Calculate Q_{solar} . Compare with same window having SC= 0.25 (low-e glass).
3	People and equipment load	Meeting room: 10 people (light work: $q_s=70\text{W}$, $q_l=60\text{W}$, CLF= 0.90). 6 laptops at 45W each ($F_{\text{use}}=0.8$, $F_{\text{load}}=0.7$). Calculate total sensible and latent internal loads.
4	Outside air ventilation load	Canteen: 20 occupants, 50 m^2 floor area. Outside: 35°C , $W=18\text{g/kg}$. Inside: 22°C , $W=9\text{g/kg}$. Calculate Q_{oa} ventilation and select ASHRAE 62.1 flow rate.
5	Complete room calculation	$8\times 5\text{m}$ bedroom, single-story house, Cairo (30°N), July, 4pm. East window ($2\text{m} \times 1.2\text{m}$, SC= 0.75). Flat concrete roof. 2 occupants at rest. Outdoor: 38°C . Indoor: 24°C . Complete full load calculation and select AC capacity.

Lesson Summary

Key Concept	Summary
7 heat gain sources	Solar glass, transmission (walls/roof), people (sensible+latent), lighting, equipment, infiltration, ventilation outdoor air. All must be evaluated.
CLTD/CLF method	Simplified approach using ASHRAE tables. CLTD accounts for time lag in walls. CLF accounts for thermal storage reducing instantaneous loads.
Transmission formula	$Q = U \times A \times \text{CLTD}$. U from construction, A from drawings, CLTD from ASHRAE tables by construction type, orientation, hour.
Solar formula	$Q = A \times \text{SC} \times \text{SHGF} \times \text{CLF}$. SC from glass spec, SHGF from solar tables, CLF from orientation/hour tables.
People heat	Sensible: $N \times q_s \times \text{CLF}$. Latent: $N \times q_l$ (no CLF). Values by activity from ASHRAE Fundamentals Table 1.
Ventilation load	$Q_s = 0.33 \times Q_{\text{oa}} \times \Delta T$ (SI). $Q_l = 0.8 \times Q_{\text{oa}} \times \Delta W$. Size Q_{oa} per ASHRAE 62.1 minimum outdoor air requirements.
Safety factor	Add 10–15% to calculated total before selecting equipment. This accounts for construction tolerances and future heat gains.
Equipment selection	Select the next standard size above (calculated load $\times$ safety factor). Never oversize by more than 25% — short-cycling reduces dehumidification.

What Is Next — HVAC-011

HVAC-011: Split AC Systems — Components of split air conditioning systems (outdoor unit, indoor unit, refrigerant circuit, controls). Types: wall-mounted, floor-standing, ceiling-cassette, ducted, and multi-split. Inverter vs fixed-speed compressors. EER and COP performance ratings. Selection criteria and installation requirements for residential and light commercial applications.

Resource	Details
HVAC PDF Series	HVAC-001 through HVAC-030 — ayetechub.com/pdfs.html
HVAC Calculator	Free Solar PV & HVAC sizing tools — ayetechub.com/downloads
Community	Telegram: t.me/ayetechub